



3.5

CUBIC SPLINE INTERPOLATION

Cubic Spline Interpolation¹

The previous sections concerned the approximation of arbitrary functions on closed intervals using a single polynomial. However, high-degree polynomials can oscillate erratically; that is, a minor fluctuation over a small portion of the interval can induce large fluctuations over the entire range. We will see a good example of this in Figure 3.14 at the end of this section.

An alternative approach is to divide the approximation interval into a collection of subintervals and construct a (generally) different approximating polynomial on each subinterval. This is called **piecewise-polynomial approximation**.

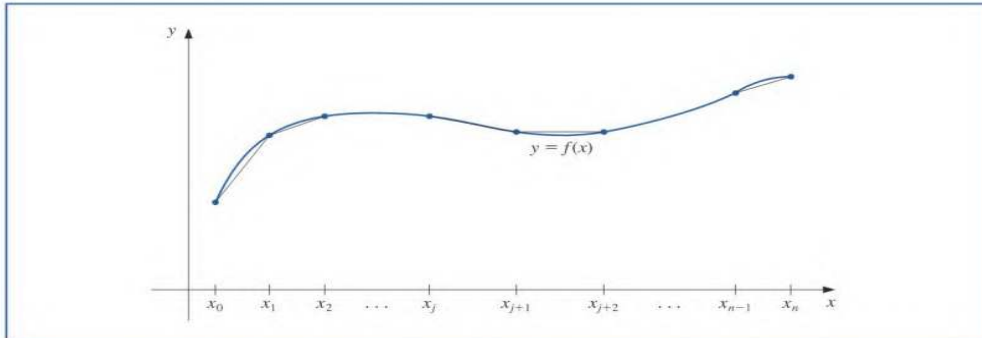
Piecewise-Polynomial Approximation

The simplest piecewise-polynomial approximation is **piecewise-linear** interpolation, which consists of joining a set of data points

$$\{(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))\}$$

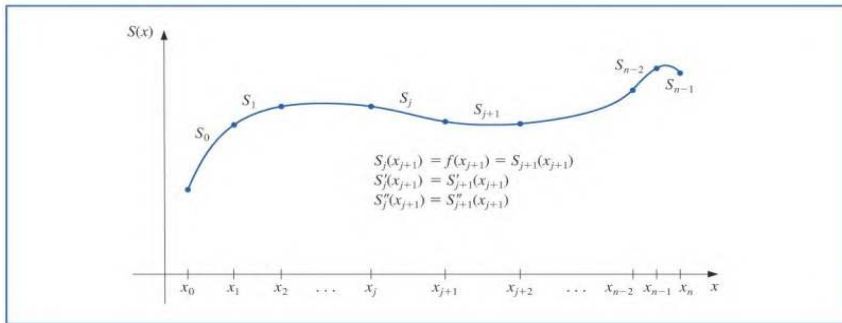
by a series of straight lines, as shown in Figure 3.7.

figure 3.7



Cubic spline
↓ ↓
Natural clamp

Figure 3.8



Definition 3.10 Given a function f defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \dots < x_n = b$, a **cubic spline interpolant** S for f is a function that satisfies the following conditions:

- (a) $S(x)$ is a cubic polynomial, denoted $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, 1, \dots, n-1$;
- (b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each $j = 0, 1, \dots, n-1$;
- (c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$; (Implied by (b).)
- (d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$;
- (e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \dots, n-2$;
- (f) One of the following sets of boundary conditions is satisfied:
 - (i) $S''(x_0) = S''(x_n) = 0$ (**natural (or free) boundary**);
 - (ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (**clamped boundary**).

Natural Splines

$$A = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ h_0 & 2(h_0 + h_1) & h_1 & \dots & 0 \\ 0 & h_1 & 2(h_1 + h_2) & \dots & h_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix},$$

and $\mathbf{b}$ and $\mathbf{x}$ are the vectors

$$\mathbf{b} = \begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ \vdots \\ \frac{3}{h_{n-1}}(a_n - a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1} - a_{n-2}) \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

$$b_0 = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0)$$

$$b_1 = \frac{1}{h_1}(a_2 - a_1) - \frac{h_1}{3}(c_2 + 2c_1)$$

$$d_0 = \frac{1}{3h_0}(c_1 - c_0)$$

$$d_1 = \frac{1}{3h_1}(c_2 - c_1)$$

for x_0, x_1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 2 \times 2$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For natural

for x_0, x_1, x_2

$$A = \begin{bmatrix} 1 & 0 & 0 \\ h_0 & 2(h_0 + h_1) & h_1 \\ 0 & h_1 & 2(h_1 + h_2) \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ 0 \end{bmatrix}$$

EXERCISE SET 3.5

- Determine the natural cubic spline S that interpolates the data $f(0) = 0$, $f(1) = 1$, and $f(2) = 2$.
- Determine the clamped cubic spline s that interpolates the data $f(0) = 0$, $f(1) = 1$ and $f(2) = 2$ and satisfies $s'(0) = s'(2) = 1$.
- Construct the natural cubic spline for the following data.

a.

x	$f(x)$
8.3	17.56492
8.6	18.50515

b.

x	$f(x)$
0.8	0.22363362
1.0	0.65809197

c.

x	$f(x)$
-0.5	-0.0247500
-0.25	0.3349375
0	1.1010000

d.

x	$f(x)$
0.1	-0.62049958
0.2	-0.28398668
0.3	0.00660095
0.4	0.24842440

- Construct the natural cubic spline for the following data.

a.

x	$f(x)$
0	1.00000
0.5	2.71828

b.

x	$f(x)$
-0.25	1.33203
0.25	0.800781

c.

x	$f(x)$
0.1	-0.29004996
0.2	-0.56079734
0.3	-0.81401972

d.

x	$f(x)$
-1	0.86199480
-0.5	0.95802009
0	1.0986123
0.5	1.2943767

- The data in Exercise 3 were generated using the following functions. Use the cubic splines constructed in Exercise 3 for the given value of x to approximate $f(x)$ and $f'(x)$ and calculate the actual error.
 - $f(x) = x \ln x$; approximate $f(8.4)$ and $f'(8.4)$.
 - $f(x) = \sin(e^x - 2)$; approximate $f(0.9)$ and $f'(0.9)$.
 - $f(x) = x^3 + 4.001x^2 + 4.002x + 1.101$; approximate $f(-\frac{1}{3})$ and $f'(-\frac{1}{3})$.
 - $f(x) = x \cos x - 2x^2 + 3x - 1$; approximate $f(0.25)$ and $f'(0.25)$.

Exo 3.5

Natural spline

3a

x	$f(x)$
$x_0 = 8.3$	$17.56492 = a_0$
$x_1 = 8.6$	$18.50515 = a_1$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$C = A^{-1} b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{matrix} c_0 \\ c_1 \end{matrix}$$

$$b_0 = \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{3} (c_1 + 2c_0)$$

$$d_0 = \left[\frac{1}{3h_0} \right] (c_1 - c_0)$$

$$h_0 = x_1 - x_0 = 8.6 - 8.3 = 0.3$$

$$b_0 = \frac{1}{0.3} (18.50515 - 17.56492) - \frac{0.3}{3} (0) = 3.1341$$

$$d_0 = \frac{1}{3(0.3)} (0) = 0$$

$$s(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \quad [x_0, x_1]$$

$$= 17.56492 + 3.1341(x - 8.3) + 0 + 0 \quad [8.3, 8.6]$$

$$= 17.56492 + 3.1341(x - 8.3) \quad [8.3, 8.6]$$

(3C)

X	f(x)
$x_0 = -0.5$	$-0.02475 = a_0$
$x_1 = -0.25$	$0.3349375 = a_1$
$x_2 = 0$	$1.101 = a_2$

$$h_0 = x_1 - x_0 = -0.25 + 0.5 = 0.25$$

$$h_1 = x_2 - x_1 = 0 + 0.25 = 0.25$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ h_0 & 2(h_0 + h_1) & h_1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 2(0.25+0.25) & 0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 1 & 0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

3x3 natural spline

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{1}{h_1}(a_2 - a_1) - \frac{2}{h_0}(a_1 - a_0) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{0.25}(1.101 - 0.3349375) - \frac{2}{0.25}(0.3349375 - (-0.02475)) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.8765 \\ 0 \\ 0 \end{bmatrix}$$

$$A^{-1}b = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4.8765 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.8765 \\ 0 \\ 0 \end{bmatrix} = \begin{matrix} c_0 \\ c_1 \\ c_2 \end{matrix}$$

$$b_0 = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0) = \frac{1}{0.25}(0.3349375 - (-0.02475)) - \frac{0.25}{3}(4.8765 + 2(0)) = 1.62375$$

$$b_1 = \frac{1}{h_1}(a_2 - a_1) - \frac{h_1}{3}(c_2 + 2c_1) = \frac{1}{0.25}(1.101 - 0.3349375) - \frac{0.25}{3}(0 + 2(4.8765)) = 2.2515$$

$$d_0 = \frac{1}{3h_0}(c_1 - c_0) = \frac{1}{3(0.25)}(4.8765 - 0) = 6.502$$

$$d_1 = \frac{1}{3h_1}(c_2 - c_1) = \frac{1}{3(0.25)}(0 - 4.8765) = -6.502$$

$$S(x) = \begin{cases} a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 & [x_0, x_1] \\ a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 & [x_1, x_2] \end{cases}$$

$$= \begin{cases} -0.02475 + 1.62375(x + 0.5) + 0 + 6.502(x + 0.5)^3 & [-0.5, -0.25] \\ 0.3349375 + 2.2515(x + 0.25) + 4.8765(x + 0.25)^2 - 6.502(x + 0.25)^3 & [-0.25, 0] \end{cases}$$

Clamped

Online

determine the linear system $Ax = b$, where

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 & \dots & 0 \\ h_0 & 2(h_0 + h_1) & h_1 & \dots & 0 \\ 0 & h_1 & 2(h_1 + h_2) & h_2 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & \dots & \dots & h_{n-2} & 2(h_{n-2} + h_{n-1}) \\ 0 & \dots & \dots & h_{n-1} & 2h_{n-1} \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ \vdots \\ \frac{3}{h_{n-1}}(a_n - a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1} - a_{n-2}) \\ 3f'(b) - \frac{3}{h_{n-1}}(a_n - a_{n-1}) \end{bmatrix}, \quad \text{and } x = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix}$$

For clamped

for x_0, x_1

$$A = \begin{bmatrix} 2h_0 & h_0 \\ h_0 & 2h_0 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ 3f'(b) - \frac{3}{h_0}(a_1 - a_0) \end{bmatrix}$$

for x_0, x_1, x_2

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 \\ h_0 & 2(h_0 + h_1) & h_1 \\ 0 & h_1 & 2h_1 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ 3f'(b) - \frac{3}{h_1}(a_2 - a_1) \end{bmatrix}$$

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3. Construct the natural cubic spline for the following data.

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5. The data in Exercise 3 were generated using the following functions. Use the cubic splines constructed in Exercise 3 for the given value of x to approximate $f(x)$ and $f'(x)$ and calculate the actual error.

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- $f(x) = x^3 + 4.001x^2 + 4.002x + 1.101$; approximate $f(-\frac{1}{3})$ and $f'(-\frac{1}{3})$.
- $f(x) = x \cos x - 2x^2 + 3x - 1$; approximate $f(0.25)$ and $f'(0.25)$.

7. Construct the clamped cubic spline using the data of Exercise 3 and the fact that

- a. $f'(8.3) = 1.116256$ and $f'(8.6) = 1.151762$.
 - b. $f'(0.8) = 2.1691753$ and $f'(1.0) = 2.0466965$.
 - c. $f'(-0.5) = 0.7510000$ and $f'(0) = 4.0020000$.
 - d. $f'(0.1) = 3.58502082$ and $f'(0.4) = 2.16529366$.
8. Construct the clamped cubic spline using the data of Exercise 3 and the fact that
- a. $f'(0) = 2$ and $f'(0.5) = 5.43656$.
 - b. $f'(-0.25) = 0.437500$ and $f'(0.25) = -0.625000$.
 - c. $f'(0.1) = -2.8004996$ and $f'(0) = -2.9734038$.
 - d. $f'(-1) = 0.15536240$ and $f'(0.5) = 0.45186276$.

11. A natural cubic spline S on $[0, 2]$ is defined by

$$S(x) = \begin{cases} S_0(x) = 1 + 2x - x^3, & \text{if } 0 \leq x < 1, \\ S_1(x) = 2 + b(x-1) + c(x-1)^2 + d(x-1)^3, & \text{if } 1 \leq x \leq 2. \end{cases}$$

Find b , c , and d .

12. A natural cubic spline S is defined by

$$S(x) = \begin{cases} S_0(x) = 1 + B(x-1) - D(x-1)^3, & \text{if } 1 \leq x < 2, \\ S_1(x) = 1 + b(x-2) - \frac{3}{4}(x-2)^2 + d(x-2)^3, & \text{if } 2 \leq x \leq 3. \end{cases}$$

If S interpolates the data $(1, 1)$, $(2, 1)$, and $(3, 0)$, find B , D , b , and d .

13. A clamped cubic spline s for a function f is defined on $[1, 3]$ by

$$s(x) = \begin{cases} s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3, & \text{if } 1 \leq x < 2, \\ s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3, & \text{if } 2 \leq x \leq 3. \end{cases}$$

Given $f'(1) = f'(3)$, find a , b , c , and d .

14. A clamped cubic spline s for a function f is defined by

$$s(x) = \begin{cases} s_0(x) = 1 + Bx + 2x^2 - 2x^3, & \text{if } 0 \leq x < 1, \\ s_1(x) = 1 + b(x-1) - 4(x-1)^2 + 7(x-1)^3, & \text{if } 1 \leq x \leq 2. \end{cases}$$

Find $f'(0)$ and $f'(2)$.

7a

x	f(x)	f'(x)
$x_0 = 8.3$	17.56492 = a_0	3.116256 = $f'(x_0)$
$x_1 = 8.6$	18.50515 = a_1	3.151762 = $f'(x_1)$

$$A = \begin{bmatrix} 2h_0 & h_0 \\ h_0 & 2h_0 \end{bmatrix} = \begin{bmatrix} 2(0.3) & 0.3 \\ 0.3 & 1(0.3) \end{bmatrix} = \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 0.6 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} \frac{20}{9} & -\frac{10}{9} \\ -\frac{10}{9} & \frac{20}{9} \end{bmatrix} \quad \text{In Fraction because of recurring decimal}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(x_0) \\ 3f'(x_1) - \frac{3}{h_0}(a_1 - a_0) \end{bmatrix} = \begin{bmatrix} \frac{3}{0.3}(18.50515 - 17.56492) - 3(3.116256) \\ 3(3.151762) - \frac{3}{0.3}(18.50515 - 17.56492) \end{bmatrix} = \begin{bmatrix} 0.053532 \\ 0.052986 \end{bmatrix}$$

$$c = A^{-1}b = \begin{bmatrix} \frac{20}{9} & -\frac{10}{9} \\ -\frac{10}{9} & \frac{20}{9} \end{bmatrix} \begin{bmatrix} 0.053532 \\ 0.052986 \end{bmatrix} = \begin{bmatrix} 0.0600867 \\ 0.0582667 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}$$

$$h_0 = x_1 - x_0 = 8.6 - 8.3 = 0.3$$

$$b_0 = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0)$$

$$b_0 = \frac{1}{0.3}(18.50515 - 17.56492) - \frac{0.3}{3}(0.0582667 - 2(0.0600867)) = 3.116256$$

$$d_0 = \frac{1}{3h_0}(c_1 - c_0)$$

$$d_0 = \frac{1}{3(0.3)}(0.0582667 - 0.0600867) = -0.00202222$$

$$s(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \quad [x_0, x_1]$$

$$= 17.56492 + 3.116256(x - 8.3) + 0.0600867(x - 8.3)^2 - 0.00202222(x - 8.3)^3 \quad [8.3, 8.6]$$

7c

X	f(x)	f'(x)
$x_0 = -0.5$	$-0.02475 = a_0$	$f'(-0.5) = 0.751 = f'(a)$
$x_1 = -0.25$	$0.3349375 = a_1$	$f'(-0.25) = 4.002 = f'(b)$
$x_2 = 0$	$1.101 = a_2$	

3x3 clamped spline

$$h_0 = x_1 - x_0 = -0.25 + 0.5 = 0.25$$

$$h_1 = x_2 - x_1 = 0 + 0.25 = 0.25$$

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 \\ h_0 & 2(h_0+h_1) & h_1 \\ 0 & h_1 & 2h_1 \end{bmatrix}$$

$$= \begin{bmatrix} 2(0.25) & 0.25 & 0 \\ 0.25 & 2(0.25+0.25) & 0.25 \\ 0 & 0.25 & 2(0.25) \end{bmatrix}$$

$$A = \begin{bmatrix} 0.5 & 0.25 & 0 \\ 0.25 & 1 & 0.25 \\ 0 & 0.25 & 0.5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 7/3 & -1/3 & 1/3 \\ -1/3 & 4/3 & -2/3 \\ 1/3 & -2/3 & 7/3 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ 3f'(b) - \frac{3}{h_1}(a_2 - a_1) \end{bmatrix} = \begin{bmatrix} \frac{3}{0.25}(0.3349375 - 0.02475) - 3(0.751) \\ \frac{3}{0.25}(1.101 - 0.3349375) - \frac{3}{0.25}(0.3349375 - 0.02475) \\ 3(4.002) - \frac{3}{0.25}(1.101 - 0.3349375) \end{bmatrix} = \begin{bmatrix} 2.06325 \\ 4.8765 \\ 2.81325 \end{bmatrix}$$

$$A^{-1}b = \begin{bmatrix} 7/3 & -1/3 & 1/3 \\ -1/3 & 4/3 & -2/3 \\ 1/3 & -2/3 & 7/3 \end{bmatrix} \begin{bmatrix} 2.06325 \\ 4.8765 \\ 2.81325 \end{bmatrix} = \begin{bmatrix} 2.501 \\ 3.251 \\ 4.001 \end{bmatrix} \begin{matrix} c_0 \\ c_1 \\ c_2 \end{matrix}$$

$$b_0 = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0) = \frac{1}{0.25}(0.3349375 - 0.02475) - \frac{0.25}{3}(3.251 - 2(2.501)) = 0.751$$

$$b_1 = \frac{1}{h_1}(a_2 - a_1) - \frac{h_1}{3}(c_2 + 2c_1) = \frac{1}{0.25}(1.101 - 0.3349375) - \frac{0.25}{3}(4.001 - 2(3.251)) = 2.189$$

$$d_0 = \frac{1}{3h_0}(c_1 - c_0) = \frac{1}{3(0.25)}(3.251 - 2.501) = 1$$

$$d_1 = \frac{1}{3h_1}(c_2 - c_1) = \frac{1}{3(0.25)}(4.001 - 3.251) = 1$$

$$S(x) = \begin{cases} a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 & [x_0, x_1] \\ a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 & [x_1, x_2] \end{cases}$$

$$= \begin{cases} -0.02475 + 0.751(x + 0.5) + 2.501(x + 0.5)^2 + (x + 0.5)^3 & [-0.5, -0.25] \\ 0.3349375 + 2.189(x + 0.25) + 3.251(x + 0.25)^2 + (x + 0.25)^3 & [-0.25, 0] \end{cases}$$

(7c)

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ 3f'(b) - \frac{3}{h_1}(a_2 - a_1) \end{bmatrix} = \begin{bmatrix} \frac{3}{0.25}(0.3349375 + 0.02475) - 3(0.75) \\ \frac{3}{0.25}(1.101 - 0.3349375) - \frac{3}{0.25}(0.3349375 + 0.02475) \\ 3(4.002) - \frac{3}{0.25}(1.101 - 0.3349375) \end{bmatrix}$$

$$\textcircled{b_1} = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0) = \frac{1}{0.25}(0.3349375 + 0.02475) - \frac{0.25}{3}(3.251 + 2(2.501)) = 0.751$$

$$\textcircled{b_2} = \frac{1}{h_1}(a_2 - a_1) - \frac{h_1}{3}(c_2 + 2c_1) = \frac{1}{0.25}(1.101 - 0.3349375) - \frac{0.25}{3}(4.001 + 2(3.251)) = 2.189$$

$$\textcircled{d_0} = \frac{1}{3h_0}(c_1 - c_0) = \frac{1}{3(0.25)}[3.251 - 2.501] = 1$$

$$\textcircled{d_1} = \frac{1}{3h_1}(c_2 - c_1) = \frac{1}{3(0.25)}[4.001 - 3.251] = 1$$

$$\textcircled{S(x)} = \begin{cases} -0.02475 + 0.751(x+0.5) + 2.501(x+0.5)^2 + (x+0.5)^3, [-0.5, -0.25] \textcircled{1} \\ -0.3349375 + 2.189(x+0.25) + 3.251(x+0.25)^2 + (x+0.25)^3, [-0.25, 0] \textcircled{2} \end{cases}$$

$$[0, 0.25] \textcircled{3}$$

$$[0.25, 0.5] \textcircled{4}$$

$$x = 0.1 \textcircled{5}$$

$$x = 0.3 \textcircled{6}$$

①

$$f(0) = 0$$

$$f(1) = 1$$

$$f(2) = 2$$

① $f(0) = 0, f(1) = 1, f(2) = 2$

x	f(x)
$x_0 = 0$	$0 = a_0$
$x_1 = 1$	$1 = a_1$
$x_2 = 2$	$2 = a_2$

$$h_0 = 1 - 0 = 1, h_1 = 2 - 1 = 1$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2(1+1) & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 4 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 0.25 & -0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{1}(2-1) - \frac{3}{1}(1-0) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A^{-1}b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}$$

$$b_0 = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0)$$

$$= \frac{1}{1}(1 - 0) - \frac{1}{3}(0 + 0) = 1$$

$$b_1 = \frac{1}{1}(2 - 1) - \frac{1}{3}(0 + 0) = 1$$

$$b_2 = \frac{1}{3h_0}(c_1 - c_0) = \frac{1}{3(1)}(0 - 0) = 0$$

$$d_1 = \frac{1}{3h_1}(c_2 - c_1) = 0$$

$$S(x) = \begin{cases} 0 + 1(x - 0) + 0 + 0, & [0, 1] \\ 1 + 1(x - 1) + 0 + 0, & [1, 2] \end{cases}$$

$$\Rightarrow S(x) = \begin{cases} x, & [0, 1] \\ 1 + (x - 1), & [1, 2] \end{cases}$$

$$\Rightarrow S(x) = x, [0, 2]$$

Exo 3.5

Natural spline

(3a) $\begin{array}{c|c} x & f(x) \\ \hline 8.3 & 17.56492 \\ 8.6 & 18.50515 \end{array}$

$$b_0 = \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{3} (c_1 + 2c_0) \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{given}$$

$$d_0 = \frac{1}{3h_0} (c_1 - c_0)$$

$$h_0 = x_1 - x_0 = 8.6 - 8.3 = 0.3$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$C = A^{-1} b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{matrix} c_0 \\ c_1 \end{matrix}$$

$$b_0 = \frac{1}{0.3} (18.50515 - 17.56492) - \frac{0.3}{3} (0) = 3.1341$$

$$d_0 = \frac{1}{3(0.3)} (0) = 0$$

given $s(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \quad [x_0, x_1]$

$$= 17.56492 + 3.1341(x - 8.3) + 0 + 0 \quad [8.3, 8.6]$$

$$= 17.56492 + 3.1341(x - 8.3) \quad [8.3, 8.6]$$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 $b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 $A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 $C = A^{-1} b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{matrix} c_0 \\ c_1 \end{matrix}$

$h_0 = x_1 - x_0 = 8.6 - 8.3 = 0.3$
 $b_0 = \frac{1}{0.3} (18.50515 - 17.56492) - \frac{0.3}{3} (0) = 3.1341$
 $d_0 = \frac{1}{3(0.3)} (0 - 0) = 0$

$s(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3, [x_0, x_1] \text{ given}$
 $= 17.56492 + 3.1341(x - 8.3) + 0 + 0, [8.3, 8.6]$
 $= 17.56492 + 3.1341(x - 8.3), [8.3, 8.6]$

3c

x	f(x)
$x_0 = -0.5$	$-0.02475 = a_0$
$x_1 = -0.25$	$0.3349375 = a_1$
$x_2 = 0$	$1.101 = a_2$

3x3 natural spline

$$h_0 = x_1 - x_0 = -0.25 + 0.5 = 0.25$$

$$h_1 = x_2 - x_1 = 0 + 0.25 = 0.25$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix} \leftarrow$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ h_0 & 2(h_0+h_1) & h_1 \\ 0 & 0 & 1 \end{bmatrix} \leftarrow$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 2(0.25+0.25) & 0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{1}{h_1}(a_2 - a_1) - \frac{2}{h_0} (a_1 - a_0) \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{0.25} (1.101 - 0.3349375) - \frac{2}{0.25} (0.3349375 - (-0.02475)) \\ 0 \end{bmatrix} = \begin{bmatrix} 4.8765 \\ 0 \end{bmatrix} \leftarrow$$

$$A^{-1} b = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & -0.25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 4.8765 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4.8765 \\ 0 \end{bmatrix} = \begin{matrix} c_0 \\ c_1 \\ c_2 \end{matrix}$$

$$b_0 = \frac{1}{h_0} (a_1 - a_0) - \frac{h_0}{2} (c_1 + 2c_0) = \frac{1}{0.25} (0.3349375 - (-0.02475)) - \frac{0.25}{2} (4.8765 + 2(0)) = 1.632375$$

$$b_1 = \frac{1}{h_1} (a_2 - a_1) - \frac{h_1}{2} (c_2 + 2c_1) = \frac{1}{0.25} (1.101 - 0.3349375) - \frac{0.25}{2} (0 + 2(4.8765)) = 2.2515$$

$$d_0 = \frac{1}{3h_0} (c_1 - c_0)$$

$$d_0 = \frac{1}{3(0.25)} (4.8765 - 0) = 6.502$$

$$d_1 = \frac{1}{3h_1} (c_2 - c_1)$$

$$= \frac{1}{3(0.25)} (0 - 4.8765) = -6.502$$

x_0, t_1, t_2

$$S(x) = \begin{cases} a_0 + b_0(x-x_0) + \frac{c_0}{2}(x-x_0)^2 + \frac{d_0}{6}(x-x_0)^3, & [x_0, x_1] \\ a_1 + b_1(x-x_1) + \frac{c_1}{2}(x-x_1)^2 + \frac{d_1}{6}(x-x_1)^3, & [x_1, x_2] \end{cases}$$

$$= \begin{cases} -0.02475 + 1.632375(x+0.5) + 0 + 6.502(x+0.5)^3, & [-0.5, -0.25] \\ 0.3349375 + 2.2515(x+0.25) + 4.8765(x+0.25)^2 - 6.502(x+0.25)^3, & [-0.25, 0] \end{cases}$$

Ex 3.5

(5a) $f(x) = x \ln x$, approx $f(8.4)$, $f'(8.4)$

from (3a) $S(x) = 17.56492 + 3.1341(x - 8.3)$ [8.3, 8.6]

approx value $S(8.4) = 17.56492 + 3.1341(8.4 - 8.3) = 17.87833$

Exact value $f(8.4) = 8.4 \ln(8.4) = 17.8771463$

$$A.E = |f(8.4) - S(8.4)| = |17.8771463 - 17.87833| \\ = 0.006867$$

$$f'(x) = 1(\ln x) + x \cdot \frac{1}{x} = \ln x + 1$$

$$f'(8.4) = \ln(8.4) + 1 = 3.128232$$

$$S'(x) = 3.1341 \Rightarrow S'(8.4) = 3.1341$$

$$|f'(8.4) - S'(8.4)| = |3.128232 - 3.1341| \\ = 0.0058$$

Ex 3.5

(5c)

$$f(x) = x^3 + 4.001x^2 + 4.002x + 1.101, \quad f(-\frac{1}{3}), \quad f(\frac{1}{3})$$

$$f(-\frac{1}{3}) = (-\frac{1}{3})^3 + 4.001(-\frac{1}{3})^2 + 4.002(-\frac{1}{3}) + 1.101$$

$$= 0.174519$$

$$f'(x) = 3x^2 + 4.001(2x) + 4.002$$

$$= 3x^2 + 8.002x + 4.002$$

$$f'(-\frac{1}{3}) = 3(-\frac{1}{3})^2 + 8.002(-\frac{1}{3}) + 4.002 = 1.668$$

$$S(x) = \begin{cases} -0.02475 + 1.032375(x + \frac{1}{2}) + 6.502(x + \frac{1}{2})^3, & [-0.5, -0.25] \\ 0.3349375 + 2.2515(x + \frac{1}{4}) + 4.8765(x + \frac{1}{4})^2 - 6.502(x + \frac{1}{4})^3, & [-0.25, 0] \end{cases}$$

$$A.E = |f'(-\frac{1}{3}) - S'(-\frac{1}{3})|$$

$$= |1.668 - 1.57421|$$

$$= 0.093$$

$$-\frac{1}{3} \in [-0.5, -0.25]$$

$$S(-\frac{1}{3}) = -0.02475 + 1.032375(-\frac{1}{3} + \frac{1}{2}) + 6.502(-\frac{1}{3} + \frac{1}{2})^3 = 0.177414$$

$$A.E = |f(-\frac{1}{3}) - S(-\frac{1}{3})| = |0.174519 - 0.177414| = 0.0028$$

$$S'(x) = 1.032375 + 6.502(3)(x + \frac{1}{2})^2$$

$$S'(-\frac{1}{3}) = 1.032375 + 6.502(3)(-\frac{1}{3} + \frac{1}{2})^2 = 1.57421$$

7c

	x	f(x)	f'(x)
x_0	-0.5	-0.02475	$f'(-0.5) = 0.751 = f'(a)$
x_1	-0.25	0.3349375	$f'(-0.25) = 4.002 = f'(b)$
x_2	0	1.101	

$h_0 = -0.25 + 0.5 = 0.25$
 $h_1 = 0 - (-0.25) = 0.25$

$$C = A^{-1}b = \begin{bmatrix} 2h_0 & h_0 & 0 \\ h_0 & 2(h_0+h_1) & h_1 \\ 0 & h_1 & 2h_1 \end{bmatrix} = \begin{bmatrix} 2(0.25) & 0.25 & 0 \\ 0.25 & 2(0.25+0.25) & 0.25 \\ 0 & 0.25 & 2(0.25) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.25 & 0 \\ 0.25 & 1 & 0.25 \\ 0 & 0.25 & 0.5 \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{1}{6}(a_1 - a_0) - \frac{3}{h_0}f'(a) \\ \frac{1}{6}(a_2 - a_1) - \frac{3}{h_1}f'(a_1) \\ 3f'(b) - \frac{3}{h_1}(a_2 - a_1) \end{bmatrix} = \begin{bmatrix} \frac{1}{6}(0.3349375 - 0.02475) - \frac{3}{0.25}(0.751) \\ \frac{1}{6}(1.101 - 0.3349375) - \frac{3}{0.25}(4.002) \\ 3(4.002) - \frac{3}{0.25}(1.101 - 0.3349375) \end{bmatrix}$$

$$= \begin{bmatrix} 2.06325 \\ 5.595875 \\ 4.345375 \end{bmatrix} \begin{matrix} b_0 \\ 4.8765 b_1 \\ 2.81325 b_2 \end{matrix}$$

$$A^{-1} = \begin{bmatrix} 7/3 & -2/3 & 1/3 \\ -2/3 & 4/3 & -2/3 \\ 1/3 & -2/3 & 7/3 \end{bmatrix}$$

$$A^{-1}b = \begin{bmatrix} 7/3 & -2/3 & 1/3 \\ -2/3 & 4/3 & -2/3 \\ 1/3 & -2/3 & 7/3 \end{bmatrix} \begin{bmatrix} 2.06325 \\ 5.595875 \\ 4.345375 \end{bmatrix} = \begin{bmatrix} 2.532125 \\ 3.18875 \\ 7.096375 \end{bmatrix} \begin{matrix} c_0 \\ c_1 \\ c_2 \end{matrix}$$

$$\begin{bmatrix} 2.501 \\ 3.251 \\ 4.001 \end{bmatrix} \Rightarrow \begin{matrix} c_1 \\ c_2 \\ c_3 \end{matrix}$$

(7c)

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ 3f'(b) - \frac{3}{h_1}(a_2 - a_1) \end{bmatrix} = \begin{bmatrix} \frac{3}{0.25}(0.3349375 + 0.02475) - 3(0.75) \\ \frac{3}{0.25}(1.101 - 0.3349375) - \frac{3}{0.25}(0.3349375 + 0.02475) \\ 3(4.002) - \frac{3}{0.25}(1.101 - 0.3349375) \end{bmatrix}$$

$$\textcircled{b_1} = \frac{1}{h_0}(a_1 - a_0) - \frac{h_0}{3}(c_1 + 2c_0) = \frac{1}{0.25}(0.3349375 + 0.02475) - \frac{0.25}{3}(3.251 + 2(2.501)) = 0.751$$

$$\textcircled{b_2} = \frac{1}{h_1}(a_2 - a_1) - \frac{h_1}{3}(c_2 + 2c_1) = \frac{1}{0.25}(1.101 - 0.3349375) - \frac{0.25}{3}(4.001 + 2(3.251)) = 2.189$$

$$\textcircled{d_0} = \frac{1}{3h_0}(c_1 - c_0) = \frac{1}{3(0.25)}[3.251 - 2.501] = 1$$

$$\textcircled{d_1} = \frac{1}{3h_1}(c_2 - c_1) = \frac{1}{3(0.25)}[4.001 - 3.251] = 1$$

$$\textcircled{S(x)} = \begin{cases} -0.02475 + 0.751(x+0.5) + 2.501(x+0.5)^2 + (x+0.5)^3, [-0.5, -0.25] \textcircled{1} \\ 0.3349375 + 2.189(x+0.25) + 3.251(x+0.25)^2 + (x+0.25)^3, [-0.25, 0] \textcircled{2} \end{cases}$$

$$[0, 0.25] \textcircled{3}$$

$$[0.25, 0.5] \textcircled{4}$$

$$x = 0.1 \textcircled{5}$$

$$x = 0.3 \textcircled{6}$$

11. A natural cubic spline S on $[0, 2]$ is defined by

$$S(x) = \begin{cases} S_0(x) = 1 + 2x - x^3, & \text{if } 0 \leq x < 1, \\ S_1(x) = 2 + b(x-1) + c(x-1)^2 + d(x-1)^3, & \text{if } 1 \leq x \leq 2. \end{cases}$$

Find b , c , and d .

$$S_0'(x) = S_1'(x) \quad 1 + 2(1) - 1^3 = 2 \xrightarrow{\text{equal}} \boxed{x=1} \\ 2 + 2c(1-1) + 2d(1-1)^2 + d(1-1)^3 = 2$$

$$S_0'(x) = 2 - 3x^2 \quad S_0'(1) = 2 - 3(1)^2 = -1 \leftarrow \text{should be equal so } b = -1 \\ S_1'(x) = b + 2c(x-1) + 3d(x-1)^2 \quad S_1'(1) = b + 2c(1-1) + 3d(1-1)^2 = b$$

$$S_0''(x) = -6x \quad S_0''(1) = -6(1) = -6 \leftarrow -6 = 2c \\ S_1''(x) = 2c + 6d(x-1) \quad S_1''(1) = 2c + 6d(1-1) = 2c \quad \begin{matrix} -6 = 2c \\ c = -3 \end{matrix} \quad b) \quad \boxed{b}$$

For natural spline $S_0''(0) = S_1''(2)$ $S_0'(a) = S_1'(1)$

$$-6(0) = 2c + 6d(2-1)$$

$$0 = 2c + 6d$$

$$0 = 2(-3) + 6d$$

$$0 = -6 + 6d$$

$$6 = 6d$$

$$\boxed{d=1}$$

13. A clamped cubic spline s for a function f is defined on $[1, 3]$ by

$$s(x) = \begin{cases} s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3, & \text{if } 1 \leq x < 2, \\ s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3, & \text{if } 2 \leq x \leq 3. \end{cases}$$

Given $f'(1) = f'(3)$, find a , b , c , and d .

$$s_0(2) = 3(2-1) + 2(2-1)^2 - (2-1)^3 = 4 \quad \leftarrow a = 4$$

$$s_1(2) = a + b(2-2) + c(2-2)^2 + d(2-2)^3 = a$$

$$s_0'(x) = 3 + 2(2)(x-1) - 3(x-1)^2$$

$$s_0'(2) = 3 + 4(2-1) - 3(2-1)^2 = 4$$

$$s_1'(x) = b + 2c(x-2) + 3d(x-2)^2$$

$$s_1'(2) = b + 2c(2-2) + 3d(2-2)^2 = b \quad \leftarrow b = 4$$

$$s_0''(x) = 4 - 3(2)(x-1)$$

$$s_0''(2) = 4 - 6(2-1) = -2 \quad \leftarrow \frac{2c}{2} = \frac{-2}{2} \quad c = -1$$

$$s_1''(x) = 2c + 3d(2)(x-2)$$

$$s_1''(2) = 2c + 6d(2-2) = 2c$$

$$s_0'(1) = s_1'(3)$$

$$3 + 4(1-1) - 3(1-1)^2 = b + 2c(3-2) + 3d(3-2)^2$$

$$3 = 4 + 2(-1) + 3d$$

$$3 = 2 + 3d$$

$$\frac{1}{3} = \frac{3d}{3} \quad d = \frac{1}{3}$$